Exercise 2

Problem 1 (Partial derivatives)**.** If a multivariate function has continuous second-order partial derivatives, then its mixed second-order partial derivatives are equal regardless of the order of differentiation. Thus, for example, the function $f(x_1, x_2)$ of two variables satisfies

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1}.$$

Consequently, it has three distinct second-order partial derivatives: $\frac{\partial^2 f}{\partial x_1^2}$, $\frac{\partial^2 f}{\partial x_1 \partial x_2}$, $\frac{\partial^2 f}{\partial x_2^2}$.

Now consider a function $g(x_1, x_2, x_3, x_4)$ of four variables with continuous fifth-order partial derivatives. How many distinct fifth-order partial derivatives does g have, counting derivatives as the same when they differ only in the order of differentiation?

Problem 2 (Poker)**.** A standard deck of cards contains 52 cards, which includes 13 ranks (2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A) and 4 suits (♡, ♠, ♢, ♣). In poker, a hand consists of 5 cards.

- In poker, *one pair* is defined as having two cards of the same rank and three other cards that have different ranks.
- *Three of a kind* consists of three cards of the same rank and two cards of different ranks.
- A *straight* consists of five consecutive ranks, but *not* a straight flush, which additionaly requires all cards to be of the same suite. An ace may be either low (A, 2, 3, 4, 5) or high (10, J, Q, K, A).

What is the probability of obtaining *two pairs*? What is the probability of achieving *three of a kind*? And what is the probability of obtaining a *straight*?

For example, {J♠, J♡, 8♢, 9♣, 10♡} is a hand with one pair, {A♠, 2♡, 8♢, 8♣, 8♡} is a three of a kind and {4♠, 5♡, 6♢, 7♣, 8♡} is a straigh.

Problem 3 (Etiquette)**.** A group of 13 friends arrives at a restaurant. To greet one another, everyone shakes hands. How many handshakes are exchanged in total?

After greetings, they proceed to their reserved *round* table. Within the group, there is a couple. The couple decided to *not* sit aside each other, to better catchup with their other friends. How many seating arrangements are possible?

Problem 4 (Secret Santa)**.** Seven people are playing Secret Santa. Each person draws one name from a hat containing all seven names. If anyone draws their own name, everyone puts their names back and the group draws again. What is the probability that the first draw is successful, with no one drawing their own name?

Problem 5 (Balls and cells)**.** Seven balls are placed into seven cells. How many of the following arrangements are there?

1. *one cell* contains *four balls*.
2. *two cells* contain either *two or three balls*, all other cells contain neither two nor three balls.

Note: The balls are indistinguishable while the cells are enumerated.